Roman Numerals & Negative Numbers

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: Write 240 as a Roman numeral.
Method & answer: 240 = 200 + 40 = CC + XL = CCXL.

Standard: What number does DCLXXIV represent?
Method & answer: D=500, C=100, LXX=70, IV=4, so DCLXXIV = 500+100+70+4 = 674.

Advanced: The temperature is -14°C. It falls by a further 9°C, then rises by 21°C. What is the final temperature?
Method & answer: -14 − 9 = -23; -23 + 21 = -2°C.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Write 150 as a Roman numeral.

2. What number does CCXV represent? A) 205 B) 215 C) 225 D) 235

3. Write the Roman numeral for 340.

4. What number does CDXX represent?

5. Which Roman numeral represents 400? A) CCCC B) CD C) DC D) LD

6. The temperature is 12°C. It falls by 19°C. What is the new temperature?

7. Write 275 as a Roman numeral.

8. Order these temperatures from coldest to warmest: -14°C, 8°C, -3°C, -22°C.

9. Select ALL of the Roman numerals below that represent numbers greater than 300: A) CCXCIV B) CCCXLI C) CDXVI D) CCLXXX E) DXX

10. What number does CCCLXV represent?

11. The temperature is -18°C. It rises by 25°C. What is the new temperature?

12. Which of these is coldest? A) -21°C B) -6°C C) 9°C D) -29°C

13. Write 490 as a Roman numeral.

14. What number does DCXLIV represent?

15. The temperature starts at -9°C and falls by 16°C. What is the new temperature?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Write 624 as a Roman numeral.

2. What number does DCCXLIX represent?

6. Order these numbers from smallest to largest: -24, 11, -40, -3.

7. What number does CMXCIX represent? A) 899 B) 989 C) 999 D) 1,009

11. Select ALL of the Roman numerals below that represent numbers between 400 and 700: A) CCCXC B) DXLIV C) DCLXXX D) CDXV E) DCCII

12. Write 456 as a Roman numeral.

3. Which Roman numeral represents 900? A) DCD B) CM C) VM D) DM

4. The temperature is 15°C. It falls by 27°C. What is the new temperature?

5. Write 888 as a Roman numeral.

8. The temperature is -28°C. It rises by 33°C. What is the new temperature?

9. Write 999 as a Roman numeral.

10. What is the difference between 14°C and -19°C?

13. What number does DCCCXCVII represent?

14. Which temperature is warmest? A) -31°C B) -8°C C) 6°C D) -40°C

15. The temperature starts at -17°C, falls by 14°C, then rises by 22°C. What is the final temperature?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Write 999 as a Roman numeral. What number is 1 more than this?

2. What number does MCDXLIV represent? A) 1,444 B) 1,344 C) 1,544 D) 1,044

3. The temperature falls from 11°C to -19°C. By how many degrees did it fall?

4. Write the Roman numeral for 749.

5. A submarine is at -85 m. It rises by 32 m, then dives a further 51 m. What is its new depth?

6. What is 400 + 90 + 9 written as a Roman numeral? A) CDXCIX B) CDIC C) DXCIX D) CDXCVIIII

7. Write 1,000 minus 1 as a Roman numeral (i.e. 999).

8. Order these temperatures from warmest to coldest: -11°C, 19°C, -36°C, 3°C.

9. Select ALL of the following that are TRUE: A) DCCXCIX represents 799 B) MCDXCII represents 1,392 C) -27°C is colder than -19°C D) -8°C is warmer than 5°C E) CDXLIV represents 444

10. What number does MCMXCIV represent? (This is a famous year.)

11. A lift starts on floor 4, goes down 11 floors, then up 5 floors. Using negative numbers for basement floors, what floor does it end on?

12. The temperature is -22°C at midnight. By 6am it has risen by 13°C, and by midday it has risen by a further 15°C. What is the midday temperature? A) 2°C B) 4°C C) 6°C D) 8°C

13. The Roman Empire era began in the year we call 27 BC. Using negative numbers for BC years (and positive for AD), write 27 BC as a negative number, then find how many years passed between this and 14 AD.

14. A number written as a Roman numeral is DCCCXLV. What is 200 less than this number?

15. The temperature at the summit of a mountain is -34°C. At its base, it is 47°C warmer. What is the temperature at the base?